

⑧ Coherent & incoherent sources :-

Coherent sources : Sources which emit waves that have same frequency and zero (or) constant phase difference, ex: Two virtual sources derived from a single source.

ex. Laser light:-



Incoherent sources : Sources which emit ^{waves} that have random frequency & phase difference

ex. Normal light,



ex: 1. light emitted by two candles

2. light emitted by two lamps

principle of Superposition of waves:-

When two or more waves overlap, the resultant displacement at any point and at any instant is the vector sum of the instantaneous displacements that would be produced that at the point by individual waves if each wave were present alone

If $y_1, y_2, \dots, y_n$ denote the displacements of n waves meeting at a point, then the resultant displacement is given by $y = y_1 + y_2 + \dots + y_n$

Case 1:- If phase difference is zero

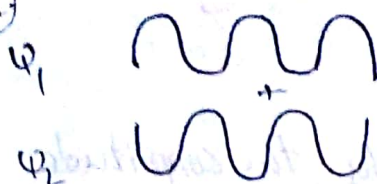


$$y = y_1 + y_2$$

It is Constructive overlap

Case 2: with phase difference

(9)



+

$$y = y_1 - y_2$$

Resultant wave
Amplitude is zero

This represents destructive overlap.

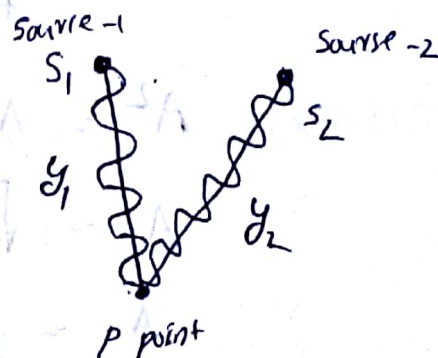
Superposition of coherent waves:-

Consider two waves travelling in space with an angular frequency ω . Let the two waves arrive at some point simultaneously.

Let y_1 and y_2 represent the displacements of two waves at this point.

$$y_1 = A_1 \sin(\omega t + \phi_1)$$

$$y_2 = A_2 \sin(\omega t + \phi_2)$$



where ϕ is phase difference.

Then according to the principle of superposition the resultant displacement at the point P is given by

$$y = y_1 + y_2 \quad \text{or} \quad y = A_1 \sin(\omega t + \phi_1) + A_2 \sin(\omega t + \phi_2)$$

$$y = A_1 [\sin \omega t \cos \phi_1 + \sin \phi_1 \cos \omega t] +$$

$$A_2 [\sin \omega t \cos \phi_2 + \cos \omega t \sin \phi_2]$$

$$y = \sin \omega t [A_1 \cos \phi_1 + A_2 \cos \phi_2] + \cos \omega t [A_1 \sin \phi_1 + A_2 \sin \phi_2]$$

$$\therefore A \cos \phi = A_1 \cos \phi_1 + A_2 \cos \phi_2 \quad \rightarrow \text{①}$$

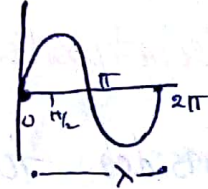
$$A \sin \phi = A_1 \sin \phi_1 + A_2 \sin \phi_2 \quad \rightarrow \text{②}$$

$$(10) y = A \cos \phi \sin \omega t + A \sin \phi \cos \omega t$$

$$= A \sin(\omega t + \phi)$$

Here A and ϕ are respectively the amplitude and initial phase of the resultant displacement

$$\boxed{\text{path difference } (\lambda) = 2\pi (\text{phase difference})}$$



Squaring and adding eq (1) & (2)

$$A^2 \cos^2 \phi + A^2 \sin^2 \phi = A_1^2 \cos^2 \phi_1 + A_2^2 \cos^2 \phi_2 + 2A_1A_2 \cos \phi_1 \cos \phi_2 + A_1^2 \sin^2 \phi_1 + A_2^2 \sin^2 \phi_2 + 2A_1A_2 \sin \phi_1 \sin \phi_2$$

$$A^2 = A_1^2 + A_2^2 + 2A_1A_2 \cos(\phi_1 - \phi_2)$$

$$A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos(\phi_1 - \phi_2)}$$

$$A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \phi} \rightarrow (3)$$

where $\phi = \phi_1 - \phi_2$ phase difference b/w two waves.

Since the intensity of a wave is proportional to square of the amplitude, the resultant intensity I of the wave from equation (3)

$$\boxed{I \propto A^2}$$

$$\boxed{I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi} \rightarrow (4)$$

where I_1 and I_2 be the intensities of the two waves

It can be seen that the amplitude (intensity) of the resultant displacement varies with phase difference of the constituent displacements.

Case 1 When $\phi = \phi_1 - \phi_2 = 0, 2\pi, 4\pi, \dots, 2n\pi$

(11) phase difference $= 2n\pi$ where $n = 0, 1, 2, \dots$

(or)

path difference $= n\lambda$ $n = 0, 1, 2, \dots$

$$\cos\phi = 1$$

$$A = A_1 + A_2 \quad \rightarrow \text{from eq (3)}$$

$$\sqrt{I} = \sqrt{I_1} + \sqrt{I_2} \quad \rightarrow \text{eq (4)}$$

Hence the resultant amplitude is the sum of the two individual amplitudes. This condition refers to the "Constructive Interference."

Case - II When $\phi = \phi_1 - \phi_2 = \pi, 3\pi, 5\pi, \dots, (2n+1)\pi$

phase difference $= (2n+1)\pi$

(or)

path diff $\lambda = \frac{(n+\frac{1}{2})\lambda}{(n+\frac{1}{2})}$ $n = 0, 1, 2, \dots$

$$\cos\phi = -1$$

$$A = |A_1 - A_2| \quad \text{and} \quad \sqrt{I} = |\sqrt{I_1} - \sqrt{I_2}|$$

Hence the resultant amplitude is the difference of the individual amplitudes and is referred to as a destructive interference.

2. Superposition of incoherent waves: Incoherent waves are the waves which do not maintain phase difference. The phase of the wave fluctuates irregularly. In case of light waves the phase fluctuates randomly at a rate of about 10^8 per second.

Light detector such as human eye, photographic films etc

- Cannot respond to such rapid changes
- (12) The detector intensity is always the average intensity

$$I_{av} = I_1 + I_2 + 2\sqrt{I_1 I_2} \langle \cos \phi \rangle$$

The average value of the $\cos \phi$ over a large time interval will be zero and hence $I_{av} = I_1 + I_2$

This implies that the superposition of incoherent waves gives uniform illumination at every point and is simply equal to the sum of the intensities of the compound waves.

Interference ;

- The variation of intensity occurs due to the redistribution of the total energy of the interfering waves is called interference.
- Interference of light is a wave phenomenon

Conditions for Steady Interference:

'The Two sources must be coherent

• The two sources must be narrow

• The two sources must be close together

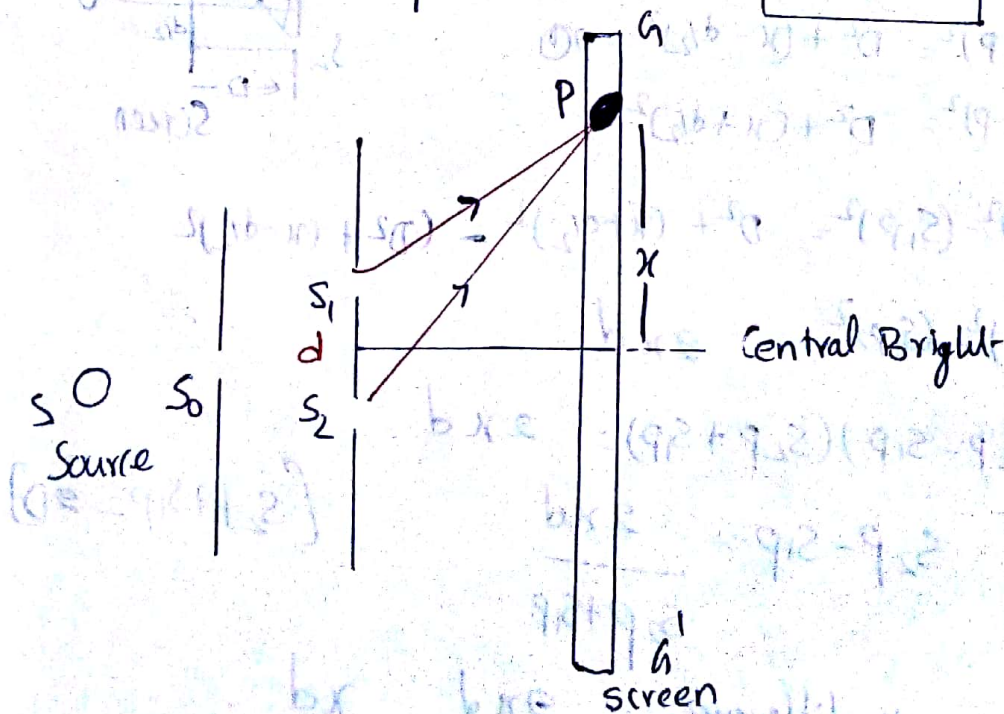
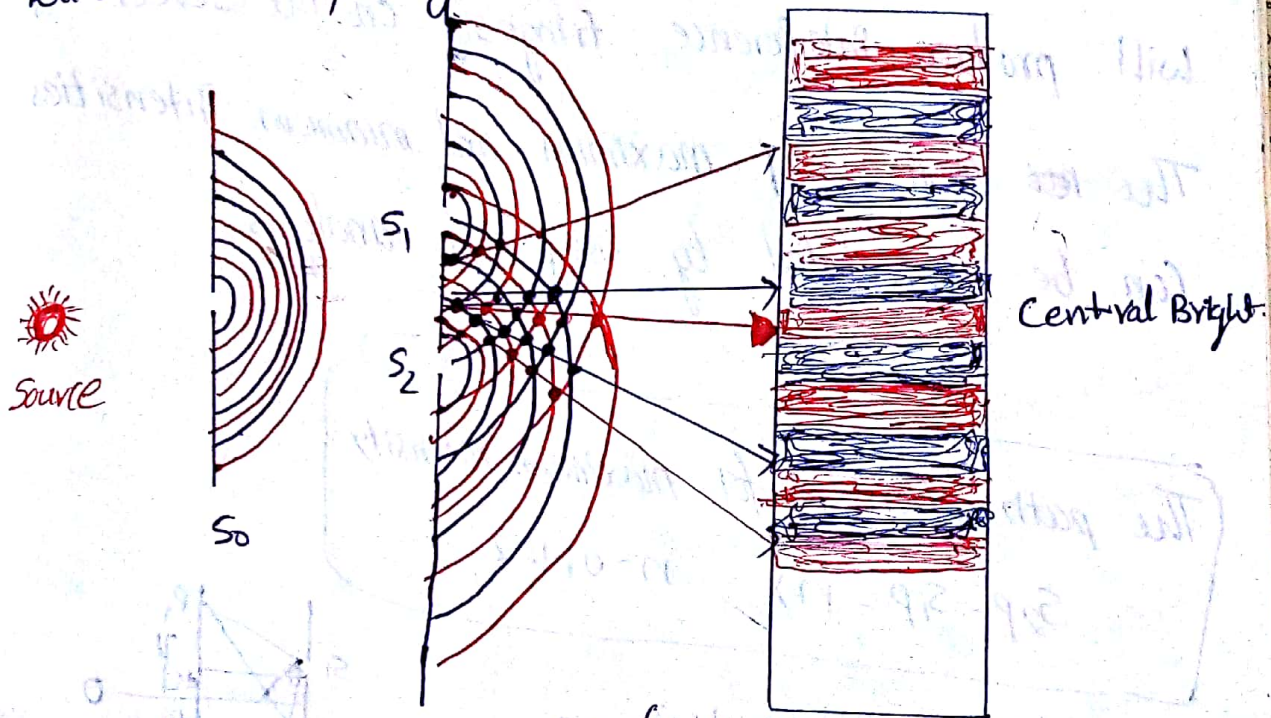
Note: The two sources must be monochromatic.

Otherwise the fringes of different colours overlap and hence interference can not be observed.

③ Young's Double Slit Experiment :-

→ Young with his experiment measured the most important characteristic of the light wave, i.e. wavelength (λ).

→ Young's experiment conclusively established the wave nature of light.



where S is source and S_0 is single slit,
and S_1, S_2 are two narrow slits,

Where, the light waves spread out from S_0 and fall on S_1 and S_2 .

The two sources S_1 and S_2 will be locked in phase. S_1 S_2 act as two coherent sources.

Thus spherical waves emanating from S_1 and S_2 will produce interference fringes on the screen.

The positions of maximum and minimum intensities can be calculated by using the analysis:

The path difference for maximum intensity

$$S_2P - S_1P = n\lambda \quad n = 0, 1, 2, \dots$$

$$(S_1P)^2 = D^2 + (x - d/2)^2 \rightarrow (1)$$

$$(S_2P)^2 = D^2 + (x + d/2)^2 \rightarrow (2)$$

$$(S_2P)^2 - (S_1P)^2 = D^2 + (x + d/2)^2 - (D^2 + (x - d/2)^2)$$

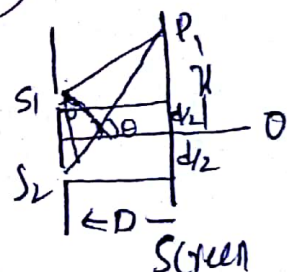
$$(S_2P)^2 - (S_1P)^2 = 2xd$$

$$(S_2P - S_1P)(S_2P + S_1P) = 2xd$$

$$S_2P - S_1P = \frac{2xd}{S_2P + S_1P}$$

$$[S_2P + S_1P = 2D]$$

$$\text{path difference} = \frac{2xd}{2D} = \frac{\lambda d}{D}$$



Bright fringes:

⑤ When source illuminates the two slits the pattern observed on the screen consists of large number of equally spaced bright and dark bands called interference fringes.

a. Bright fringes: Bright fringes occur whenever the wave from S_1 and S_2 interfere constructively i.e. on reaching P the waves with crest [or trough] superimpose at the same time and they are said to be in phase.

The condition for finding a bright fringe at P is that $S_2P - S_1P = n\lambda$

$$\frac{x_d}{D} = n\lambda$$

for n th maximum intensity position x_n

$$\frac{x_n D}{D} = n\lambda$$

where $n = 0, 1, 2, 3, \dots$

$$x_n = \frac{n\lambda D}{d}$$

$$x_n = \frac{n\lambda D}{d}$$

The ~~zero~~ bright fringes corresponding to $n=0$, is called the zero-order fringe or central maximum.

It means at P the fringe with zero path difference between two waves, on reaching the point P .

for $n=1$, is called first order bright fringe.

1.e If the path difference between the two waves on reaching P is ' λ '. Similarly $n=2$, Second order bright fringe is located where the path difference is 2λ and so on

for amplitude $A^2 = A_1^2 + A_2^2 + 2A_1A_2\cos\phi$

If amplitude of two waves is same $A_1 = A_2 = a$

$$A^2 = a^2 + a^2 + 2a^2\cos\phi$$

$$= 2a^2(1 + \cos\phi)$$

$$= 2a^2 \cdot 2\cos^2\phi/2$$

$$A^2 = 4a^2\cos^2\phi/2$$

$$\boxed{A = 2a\cos\phi/2}$$

for Intensity of ~~Bright~~ fringes

$$I \propto A^2$$

$$I = 4a^2\cos^2\phi/2$$

$a^2 = I_0$ is Initial Intensity

$$\boxed{I = 4I_0\cos^2\phi/2}$$

for maximum Intensity $\cos\phi/2 = 1$

i.e $\phi/2 = 0, \pm 2\pi, \pm 4\pi, \pm 6\pi, \dots$

(or) phase difference b/w the waves $\phi = \pm 2n\pi$

with $n = 0, 1, 2, \dots$

The corresponding path difference $\Delta x = n\lambda$

Hence maximum Intensity $\boxed{I_{\max} = 4I_0}$

Dark fringes:

(18) Dark fringes occur whenever the waves from S_1 and S_2 interfere destructively. i.e. on reaching 'P' one wave with its crest and another wave with its trough superimpose. Then the phase difference between the waves is π and the waves are said to be in opposite phase.

Destructive interference occurs at P, if S_1P and S_2P differ by a odd integral multiple of $\frac{\lambda}{2}$. Thus the condition for finding dark fringe at P is

$$\text{that } S_2P - S_1P = (2n-1)\frac{\lambda}{2} \quad n = \pm 1, \pm 2, \dots$$

$$\text{phase difference} = (2n-1)\pi$$

n is called order of dark fringe, Hence for n^{th} order dark,

$$\text{path difference} = \frac{y_n d}{D}$$

$$\frac{y_n d}{D} = (2n-1)\frac{\lambda}{2}$$

$$y_n = \frac{(2n-1)\lambda D}{2d}$$

Where y_n is the position of n^{th} minima from.

The first dark fringe

$$S_2P - S_1P = \frac{\lambda}{2} \text{ This is called first order dark } n=1,$$

$$S_2P - S_1P = \frac{3\lambda}{2} \text{ 2nd order dark.}$$

$$I = 4I_0 \cos^2\left(\frac{\phi}{2}\right)$$

⑧ minimum Intensity

$$\cos \phi/2 = 0$$

$$\therefore \phi/2 = \pm \frac{\pi}{2}, \pm \frac{3\pi}{2}, \pm \frac{5\pi}{2} \dots$$

$$\phi = \pm \pi, \pm 3\pi, \pm 5\pi$$

$$\phi = \pm (2n-1)\pi \text{ with } n = 1, 2, 3 \dots$$

$$\Delta x = (2n-1) \frac{\lambda}{2}$$

$$I_{\min} = 0.$$

$$\begin{cases} \frac{2\pi}{\lambda} \Delta x = (2n-1)\pi \\ \Delta x = (2n-1) \frac{\lambda}{2} \end{cases}$$

⇒ Fringe width β :

The distance between two adjacent bright [or dark] fringes is called the fringes width. It is denoted by β ,

The n^{th} order bright fringe occurs from the

$$\text{Central maximum at } x_n = \frac{n\lambda D}{d}$$

The $(n+1)^{\text{th}}$ order bright fringe occurs from the Central maximum $x_{n+1} = \frac{(n+1)\lambda D}{d}$

$$\beta = x_{n+1} - x_n = \boxed{\frac{\lambda D}{d} = \beta}$$

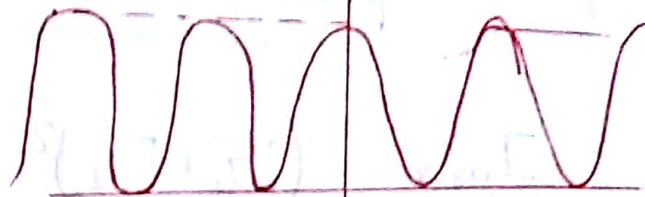
In a similar way, the same result will be obtained for the dark fringes also

$$\text{fringe width } \beta = \frac{\lambda D}{d}$$

Thus fringes width is same every where on the screen

Width of Bright fringe is equal to the width of dark fringe

$$\beta_{\text{Bright}} = \beta_{\text{dark}} = \frac{\lambda D}{d}$$



Note:-

Constructive Interference:-

(1) If the phase difference is $\phi = 2n\pi$ [every multiple of π] where $n = 0, 1, 2, 3, \dots$

when $\phi = 0, 2\pi, 4\pi, \dots, 2n\pi$

(2) If path difference $x = 2n\left(\frac{\lambda}{2}\right)$ [even multiple of half wavelength]

i.e. $x = 0, \lambda, 2\lambda, \dots, n\lambda$

The amplitude and intensity are maximum

$$A_{\text{max}} = A_1 + A_2$$

Intensity $I_{\text{max}} = (\sqrt{I_1} + \sqrt{I_2})^2 = (A_1 + A_2)^2$

If $A_1 = A_2 = a$ then

$$A_{\text{max}} = 2a$$

$$I_{\text{max}} = 4I_0$$

Destructive Interference:-

(1) If the phase difference $\phi = (2n-1)\pi$ [odd multiple of π]

when $\phi = \pi, 3\pi, 5\pi, \dots, (2n-1)\pi$

(2) If the path difference $x = (2n-1)\frac{\lambda}{2}$ [odd multiple of $\frac{\lambda}{2}$]

⑩

When $y = \frac{\lambda}{2}, \frac{3\lambda}{2}, \frac{5\lambda}{2}, \dots$

$$A_{\min} = A_1 - A_2 \quad I_{\min} = (\sqrt{I_1} - \sqrt{I_2})^2$$

If $A_1 = A_2 = A$ $A_{\min} = 0 \quad I_{\min} = 0$

$$\frac{I_{\max}}{I_{\min}} = \frac{(\sqrt{I_1} + \sqrt{I_2})^2}{(\sqrt{I_1} - \sqrt{I_2})^2} = \frac{(A_1 + A_2)^2}{(A_1 - A_2)^2}$$

$\Rightarrow$ phase difference $= \frac{2\pi}{\lambda} \times \text{path difference}$

$\phi = \frac{2\pi}{\lambda} \Delta x$
 Δx : path difference

Points:-

1. $\beta = \frac{\lambda D}{d} \Rightarrow \beta \propto \lambda$

$\beta_{\text{red}} > \beta_{\text{violet}}$ as $\lambda_{\text{red}} > \lambda_{\text{violet}}$

2. In YDSE, if blue light is used instead of red light then β decreases [$\because \lambda_B < \lambda_R$]

3. If YDSE is conducted in vacuum instead of air then β increases [$\because \lambda_{\text{vacuum}} > \lambda_{\text{air}}$]

4. In central field of view on the screen, if n_1 fringes are formed when light of wavelength λ_1 is used $\rightarrow n_2$ fringes for λ_2 wavelength.

then $x_n = \frac{n\lambda D}{d}$ const $n\lambda = \text{const}$

②

$$n_1 \lambda_2 = n_2 \lambda_1$$

• The maximum number of ~~complete~~
complete maxima on the screen
 $2n + 1$

5. The distance of n^{th} bright fringe from central maximum

$$x_{n \text{ bri}} = \frac{n\lambda D}{d} = n\beta$$

The distance of m^{th} ~~bright~~ dark fringe from central

$$\text{maximum } (x_n)_{\text{dark}} = \frac{(2m-1)}{2} \frac{\lambda D}{d} = \left(\frac{2m-1}{2}\right)\beta$$

∴ The distance between n^{th} bright and m^{th} dark fringes

$$x_{n \text{ bri}} - x_{m \text{ dar}} = n\beta - \left(\frac{2m-1}{2}\right)\beta$$

6. When white light is used YDSE the interference patterns due to different component colours of white light overlap (incoherently). The central bright fringes for different colours are the same position. Therefore, the central fringe is

white. For a point P for which $S_2P - S_1P = \frac{\lambda_b}{2}$ where

$\lambda_b = 4000 \text{ \AA}$. represents the wavelength for the blue colour. The blue component will be absent and the fringe will appear red in colour.

Slightly further where $S_2Q - S_1Q = \frac{\lambda_r}{2}$ where $\lambda_r = 7000 \text{ \AA}$ is the wavelength for the red colour, the fringes will be predominantly blue. Thus, the fringes closest on either side of the central white fringe is red and farther blue.

② Diffraction :-

The bending of light around edges of an obstacle on the encroachment of light within geometrical shadow is known as "diffraction of light".

• Diffraction is a characteristic wave property.

• Diffraction is an effect exhibited by all electromagnetic waves, water wave & sound waves.

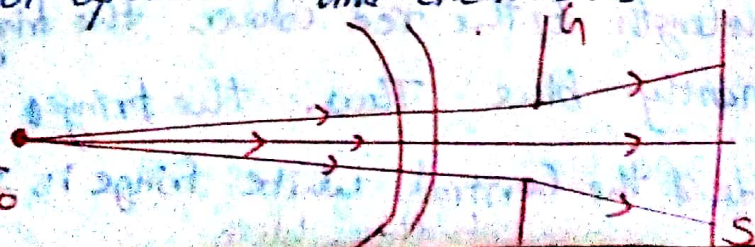
• Diffraction takes place with very small moving particles, such as atoms, neutrons, and electrons, which show wave-like properties.

• When light passes through a narrow aperture some light is found to be encroached into shadow regions.

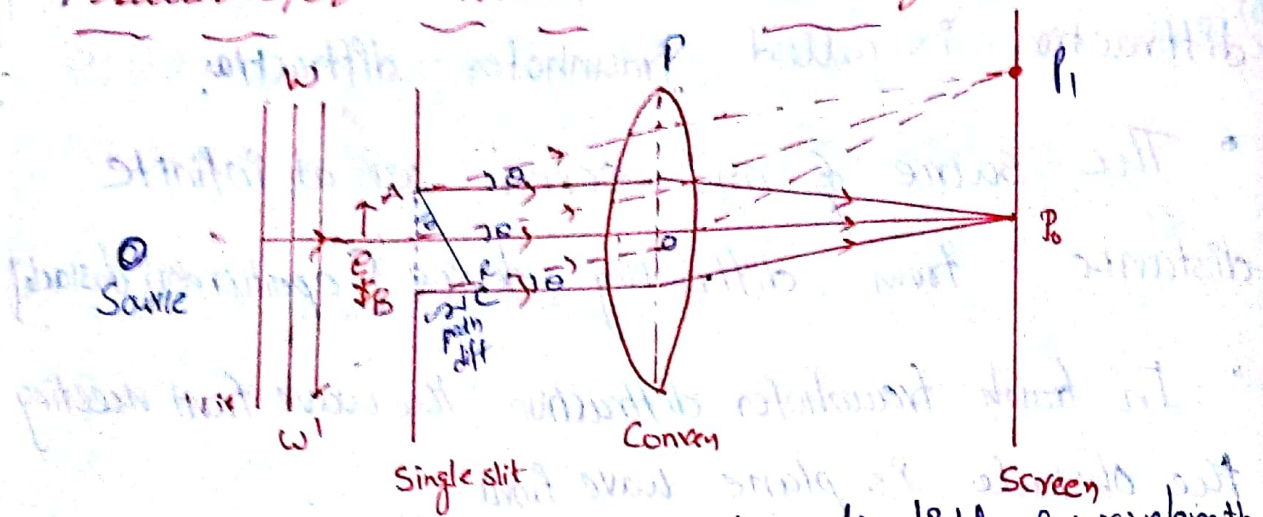
• When slit width is larger, the encroachment of light is small and negligible.

• When slit width is ~~large~~ comparable to wavelength of light the encroachment of light is more.

* If the size of the obstacle or aperture is comparable with the wavelength of light, light deviates from rectilinear propagation near edges of obstacle or aperture and encroaches into geometrical shadow.



Fraunhofer Diffraction at Single slit: 26



A plane wavefront w of monochromatic light of wavelength λ incident on narrow slit AB, of width 'e'.

According to Huygen's principle, every point of the wavefront in the plane of the slit is a source of secondary spherical wavelets, which spread out to the right in all directions. The secondary wavelets traveling normally to the slit i.e., along the direction O.P. are brought to focus at P_0 , by the Convex lens. Thus P_0 is a bright central maximum. [central fringe]

The secondary wavelets travelling at an angle θ with the normal are focused at a point P_1 on the screen.

The point P_1 is of the minimum intensity (or) maximum intensity depending upon the path difference between the secondary waves originating from the corresponding points of wave front.

Find

Find Intensity at P_1 :-

Draw a perpendicular AC on BR the path difference between Secondary wavelets from A and B is direction $\theta = BC$

$$\text{path difference} = AB \sin \theta = e \sin \theta.$$

$$\text{and corresponding phase difference} = \frac{2\pi}{\lambda} \text{ path diff} = \frac{2\pi}{\lambda} e \sin \theta.$$

let us consider that the width of the slit is divided into 'n' equal parts. and the amplitude of the wave from each part is 'a'. The phase difference between any two consecutive waves from these parts would be

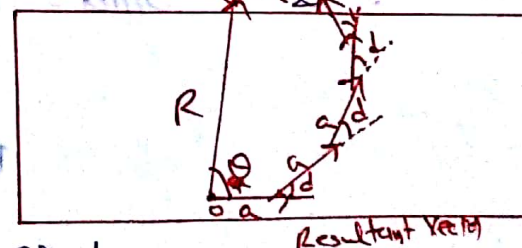
$$\Rightarrow \frac{\text{total phase diff}}{n} = \frac{1}{n} \left[\frac{2\pi}{\lambda} e \sin \theta \right] = d \text{ (say)}$$

Using the method of Vector Addition of amplitudes.

The resultant amplitude R is given by

$$R = a \frac{\sin\left(\frac{nd}{2}\right)}{\sin\left(d/2\right)} = a \frac{\sin\left(n \left[\frac{1}{n} \frac{2\pi}{\lambda} e \sin \theta \right] \right)}{\sin\left[\frac{1}{n} \frac{2\pi}{\lambda} e \sin \theta \right]}$$

$$R = a \frac{\sin\left(\frac{\pi e \sin \theta}{\lambda}\right)}{\sin\left(\frac{\pi e \sin \theta}{n \lambda}\right)}$$



Let $\alpha = \frac{\pi e \sin \theta}{\lambda}$

$$R = a \frac{\sin \alpha}{\sin(\alpha/n)}$$

Since $\frac{\alpha}{n}$ is very small

$$\text{then } \sin \frac{\alpha}{n} = \frac{\alpha}{n} \Rightarrow R = \frac{na \sin \alpha}{\alpha}$$

$$R = A \frac{\sin \alpha}{\alpha} \rightarrow \textcircled{1}$$

Now the Intensity is given by

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$$I = R^2 = A^2 \left[\frac{\sin \alpha}{\alpha} \right]^2$$

$$I = I_0 \left[\frac{\sin \alpha}{\alpha} \right]^2 \rightarrow (1)$$

Condition for maximum:-

$$R = A \frac{\sin \alpha}{\alpha} = \frac{A}{\alpha} \left[\alpha - \frac{\alpha^3}{3!} + \frac{\alpha^5}{5!} - \frac{\alpha^7}{7!} + \dots \right]$$

$$R = A \left[1 - \frac{\alpha^2}{3!} + \frac{\alpha^4}{5!} - \frac{\alpha^6}{7!} + \dots \right]$$

The value of R will be maximum where $\alpha = 0$.

$$\alpha = \frac{\pi \sin \theta}{\lambda} \Rightarrow \sin \theta = 0$$

$$\boxed{\theta = 0}$$

The condition $\theta = 0$ means that this maximum is formed by those secondary waves which travel normally to the slit,

Condition for minimum:-

The Intensity will be minimum when $\sin \alpha = 0$

$$R = \frac{A}{\alpha} \sin \alpha$$

$$\sin \alpha = \sin \left(\frac{m\pi}{2} \right) \text{ where } m = 1, 2, 3, \dots$$

$$\alpha = \pm \pi, \pm 2\pi, \pm 3\pi, \dots$$

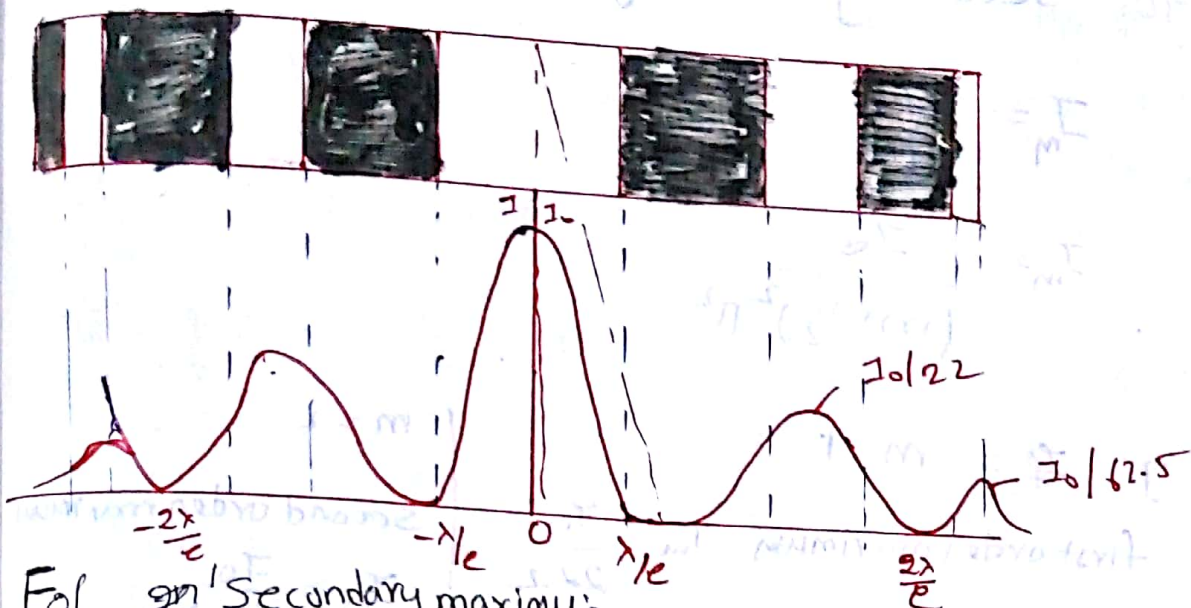
$$\frac{\pi \sin \theta}{\lambda} = \pm m\pi$$

$$\boxed{\sin \theta = \pm m\lambda}$$

where $m = 1, 2, 3, \dots$

If the path difference $\Delta = (2m) \frac{\lambda}{2}$ where $m = 1, 2, 3, \dots$ then secondary minimum are obtained at point P_1 .

Intensity distribution:-



For ^{2nd} Secondary maximum:-

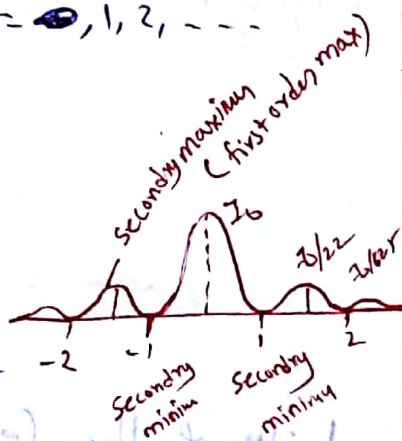
If the path difference is $(2m+1)\lambda/2$ then Secondary maxima is obtained where $m = 0, 1, 2, \dots$

path difference $\Delta = (2m+1)\lambda/2$

phase difference $\delta = \frac{2\pi}{\lambda} \text{ path diff}$

$$= \frac{2\pi}{\lambda} (2m+1)\lambda/2$$

$$\delta = 2\pi(m+1/2)$$



for maximum Intensity $I = I_0 \left[\frac{\sin \alpha}{\alpha} \right]^2$

We need 2nd secondary maximum $\sin \alpha = 1$

for $\alpha = \frac{\pi \sin \theta}{\lambda}$

Phase difference = $\frac{2\pi \sin \theta}{\lambda}$ [from Experiment]

$$2\pi(m+1/2) = \frac{2\pi \sin \theta}{\lambda}$$

$$\alpha = (m+1/2)\pi$$

f) Secondary Intensity (or) first order maximum is

$$I_m = \frac{I_0}{\alpha^2}$$

$$I_m = \frac{I_0}{(m + \frac{1}{2})^2 \pi^2}$$

$$I \text{ If } m=1$$

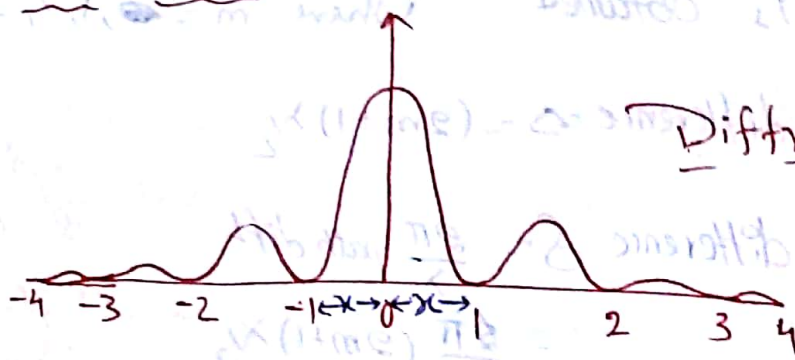
$$\text{first order maximum } I_m = \frac{I_0}{22.2}$$

$$m=2$$

second order maximum

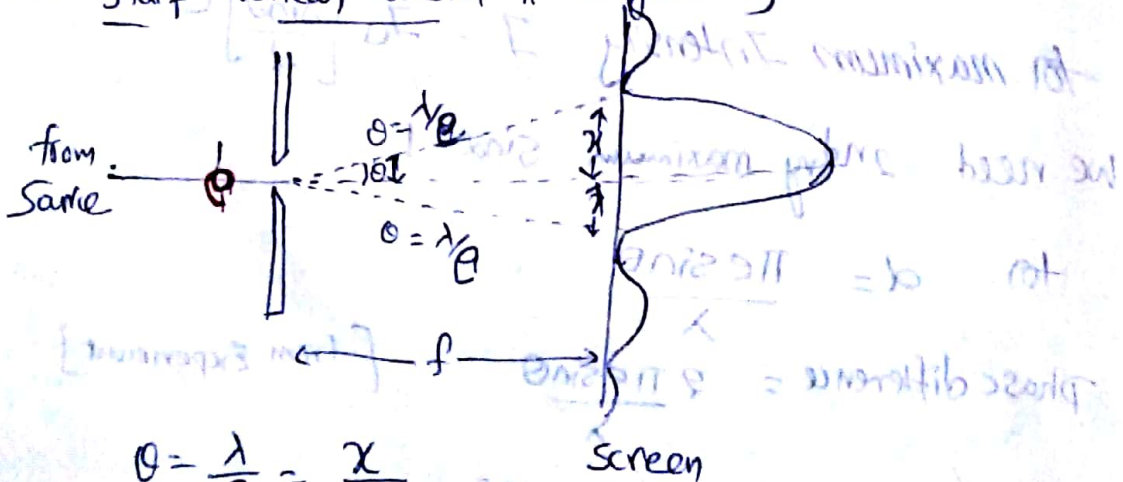
$$I_m = \frac{I_0}{62.5}$$

Angular width ω_a :-



Width of the Central maximum is

(a) Half linear width x is given by



$$\theta = \frac{\lambda}{a} = \frac{x}{f}$$

$$x = \frac{f\lambda}{a}$$

b) linear width W or $W = 2x$, i.e. The distance between two secondary minima formed on two sides of central maximum is known as the width of central maximum.

its value is given by $W = 2x = \frac{2f\lambda}{a}$

where a = slit width
 f = focal length of convex lens, L_2
 λ = wavelength of monochromatic light

Then angular width $W_\theta = \frac{\text{Linear width}}{\text{focal length}}$

$$W_\theta = \frac{W}{f} = \frac{2x}{f} = \frac{2\lambda}{a}$$

$$W_\theta = \frac{2\lambda}{a}$$

**

$$\rightarrow \sin\theta = \tan\theta = \frac{x}{f} = \frac{x}{D}$$

$$\rightarrow a \sin\theta = m\lambda \quad \text{for minimum intensity}$$

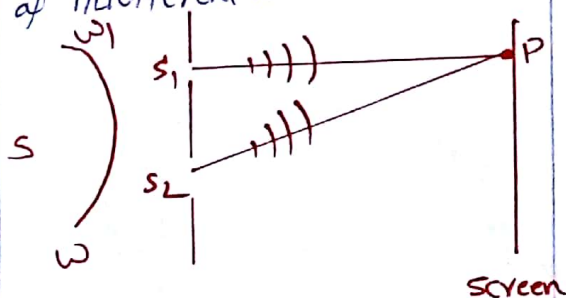
$$\rightarrow a \sin\theta = (2m+1)\frac{\lambda}{2} \text{ or } (2m-1)\frac{\lambda}{2} \quad \text{for max intensity}$$

$$\text{Linear width, } W = 2x = \frac{2f\lambda}{a}$$

$$\text{fringe width } \propto \lambda \propto \frac{1}{a}$$

Interference

- The phenomenon resulting from the superposition of secondary wavelets originating from two coherent sources is known as interference.



- Interference fringes are of equal width in case of a monochromatic source whereas these are of unequal width in case of white light.

- All bright fringes are of equal intensity.

- The intensity of all dark fringes is zero, i.e. all dark fringes are perfectly black.

Condition for maxima

$$\text{path diff } \Delta = 2n\left(\frac{\lambda}{2}\right)$$

$$\text{phase diff } \delta = 2n\pi, n=0,1,2$$

$$\text{Time diff } \theta = 2n \cdot \frac{T}{2}$$

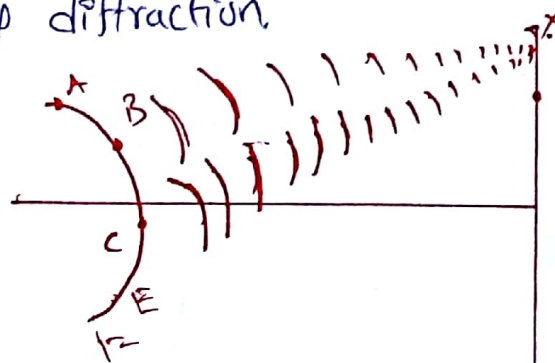
Condition for minima

$$\Delta = (2n+1) \frac{\lambda}{2}$$

$$\delta = (2n+1)\pi$$

Diffraction

The phenomenon resulting from the superposition of secondary wavelets originating from various parts of a single coherent source is known as diffraction.



Diffraction fringes are never of equal width.

The intensity of all bright fringes is not same.

The intensity of dark fringes is not zero, i.e. the dark fringes are not perfectly black.

Condition for maxima

$$\Delta = (2m+1) \frac{\lambda}{2}$$

$$\delta = (2n+1)\pi$$

$$\theta = (2n-1) \frac{T}{2}$$

Condition for min

$$\Delta = 2n \frac{\lambda}{2}$$

$$\delta = 2n\pi$$